

backoff to improve initial latency when congestion is not the cause of packet drops, and then transitions to exponential backoff to provide convergence when the network is congested. If a positive acknowledgment is received before the retransmission timer expires, the retransmission timer is stopped. Otherwise, if the retransmission timer expires, the message is retransmitted and the timer started again.

The following table illustrates minimum, maximum, and cumulative retransmission times using default parameters.

Table 21. Example MRP Retransmission Times

Metric	Transmission Time [ms]				
	330	330	528	845	1352
Min Jitter	330	330	528	845	1352
Max Jitter	413	413	660	1056	1690
Min Total	330	660	1188	2033	3385
Max Total	413	825	1485	2541	4231
Transmission #	0	1	2	3	4

The sender SHOULD initiate [Section 4.3.2, “Operational Discovery”](#) in parallel with the first retry to re-resolve the address of the destination Node if the initial transmission fails after one expected round trip. The sender SHOULD use the latest MRP parameters for the destination that result from subsequent Operational Discovery.

When a client is communicating with an ICD (i.e. the [ICD Management cluster](#) is present), the sender SHALL initiate [Section 4.3.2, “Operational Discovery”](#) in parallel with the first retry to re-resolve the address of the destination Node if the initial transmission fails after one expected round trip. The sender SHALL use the latest MRP parameters for the destination that result from subsequent Operational Discovery.

4.12.2.2. Acknowledgements

A receiver SHALL acknowledge a reliable message by either using a "piggybacked" acknowledgment in the next message destined to the peer, or a standalone acknowledgment, or both.

The acknowledgement message SHALL set the *Acknowledged Message Counter* field to the value of the *Message Counter* of the reliable message to be acknowledged.

Piggybacking Acknowledgments on Responses

Acknowledgements MAY be conveyed at the same time (i.e. piggybacked) as data in a response message. The receiver tries to optimize message transmission by deferring acknowledgments when a reliable message is received (see [Section 4.12.5.2.2, “Standalone acknowledgement processing”](#)) and piggybacking outstanding acknowledgments on messages that it needs to send back (see [Section 4.12.5.1.1, “Piggyback acknowledgment processing”](#) for more details).

Duplicate Message Detection

Since the reliable messaging protocol has a provision for the sender to retransmit messages, there